

MIDTERM EXAM

ECON301—ECONOMETRICS I

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PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

1. Aknur wants to estimate a simple regression explaining the final score (*finalscore*) in ECON301 in terms of the percentage of classes attended (*percentattend*). She has data on these two variables for a cross-section of 680 NU students that have taken ECON301 over the years. Both variables are on a scale of 0 to 100. The results from her estimation as well as some sample statistics are presented below.¹

	Estimate	Std. Error
<i>Intercept</i>	56.82479	2.19227
<i>percentattend</i>	0.09672	0.02627

$$n = 680, \quad \hat{\sigma} = 11.67, \quad \overline{finalscore} = 64.72794, \quad \overline{percentattend} = 81.70956, \\ \sum_{i=1}^{680} percentattend_i^2 = 4737305, \quad \widehat{\text{Var}(\hat{\beta}_1 | percentattend)} = \begin{bmatrix} 4.80604546 & -0.0563687409 \\ -0.0563687409 & 0.0006898672 \end{bmatrix}$$

Suppose that all assumptions, 1–5, from the conditional approach presented in lecture 3 hold.

- (a) (2 points) What is the estimated ceteris paribus effect of the percentage of classes attended on the final score? How do you interpret it?
- (b) (5 points) For the estimated ceteris paribus effect in (a), is it statistically significant? Test this hypothesis at the 5% level. Make sure to properly state and construct your hypothesis test. Irrespective of the statistical significance of the effect of attendance on final score, is this effect of *practical* significance?
- (c) (8 points) Suppose that Aknur wants to test the following hypothesis H_0 : “The effect of an increase of 20 percentage points in attendance is an increase of 2 points in the final score” versus H_a : “The effect of an increase of 20 percentage points in attendance is *less than* 2 points in the final score.” Construct this hypothesis test and test it at the 1% level.
- (d) (2 points) What is the predicted final score for the mean percentage of attendance?
- (e) (2 points) What is the estimated conditional variance of *finalscore*?
- (f) (2 points) There are students in the cross-section with 100% of attendance. What is their predicted final score?
- (g) (8 points) Provide a formula for $E(\hat{\beta}_0 + 100\hat{\beta}_1 | percentattend)$ and $\text{Var}(\hat{\beta}_0 + 100\hat{\beta}_1 | percentattend)$. Then, using the data in the display above, provide an *estimate* of the previous two moments.
- (h) (5 points) Using the results from the previous two items, propose a method to construct a 95% confidence interval for the predicted final score of a student with an attendance percentage of 100, and then use your proposed method to construct it.
- (i) (8 points) Aknur thinks that attendance does not provide the complete picture in explaining the final score in ECON301. She thinks that she should include a variable that measures *engagement with the class*. In the data, she was able to find the variable *hmkpercent*, which is the percentage of homework turned in, measured on a scale of 0 to 100. Thus, Aknur thinks that instead of the current model with which she has been working, $E[finalscore | percentattend] = \beta_0 + \beta_1 percentattend$, the correct model to estimate is

$$E[finalscore | percentattend, hmkpercent] = \delta_0 + \delta_1 percentattend + \delta_2 hmkpercent.$$

In light of this, what is the omitted variable bias when estimating the model that includes only *percentattend*? What do you think is the direction of the bias?

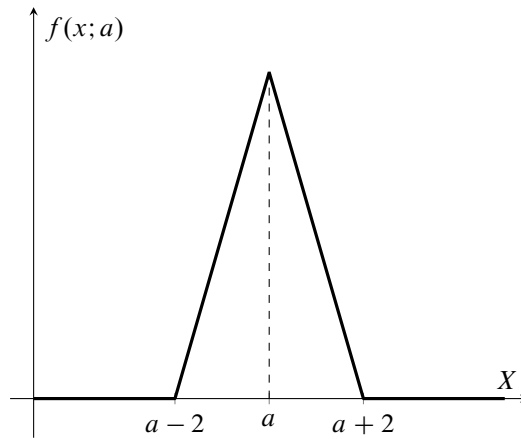
- (j) (8 points) Comment on Aknur’s work: do you think her models incorporate all variables required to explain final score? What other variables would you include and why?

¹Remember that bars above variables represent the sample mean, and the “hat” symbol $\hat{x}$ over a variable means that it is an estimate of the true value.

2. The daily tonnage of garbage handled by NU-USM is represented as the outcome of a random variable having a triangular distribution, as follows:

$$X \sim f(x; a) = \frac{1}{4}(2 - a + x)I_{[a-2, a]}(x) + \frac{1}{4}(2 + a - x)I_{(a, a+2]}(x).$$

The figure below represents graphically the distribution above. You also have the following moments from the triangular distribution: $E(X) = a$ and $E(X^2) = \frac{2}{3} + a^2$.



USM is in the process of analyzing whether or not they need to expand their facilities. It wants an estimate of the expected daily tonnage that it handles. It has collected 5 years of daily observations on tonnage handled ($n = 3,650$ observations) and provides you with the following summary statistic:

$$\sum_{i=1}^n x_i = 29,200$$

You may treat the observations as outcomes of i.i.d. random variables.

- (10 points) Show that the sample mean $\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i$ is the *least squares estimator when using only an intercept* for predicting X , that is, show that $\bar{x}$ is the solution to the problem of minimizing the sum of squares $SS(c) = \sum_{i=1}^n (x_i - c)^2$.
- (10 points) Consider the sequence of sample means $(\bar{X}_n)_{n=1}^\infty$. Does $\bar{X}_n$ converge in mean-square? Does it converge in probability? To what value does it converge to? Use these results to argue that the sample mean is the appropriate estimator to use in order to estimate the unknown parameter a from the distribution of X .
- (10 points) Define an asymptotic distribution for $\bar{X}_n$. You should be able to identify a *numerical value* for the variance of the asymptotic distribution.
- (10 points) Using the asymptotic distribution, how probable is that the outcome of $\bar{X}_n$ will be within 0.05 tons of the *actual* expected value of daily garbage tonnage handled?
- (10 points) Using your estimate of the parameter a , estimate the density function, and compute the probability that tonnage handled by USM will exceed 21 tons on any given day?

STUDENT'S t PERCENTAGE POINTS (ν = degrees of freedom)

ν	60.0%	66.7%	75.0%	80.0%	87.5%	90.0%	95.0%	97.5%	99.0%	99.5%	99.9%
1	0.325	0.577	1.000	1.376	2.414	3.078	6.314	12.706	31.821	63.657	318.31
2	0.289	0.500	0.816	1.061	1.604	1.886	2.920	4.303	6.965	9.925	22.327
3	0.277	0.476	0.765	0.978	1.423	1.638	2.353	3.182	4.541	5.841	10.215
4	0.271	0.464	0.741	0.941	1.344	1.533	2.132	2.776	3.747	4.604	7.173
5	0.267	0.457	0.727	0.920	1.301	1.476	2.015	2.571	3.365	4.032	5.893
6	0.265	0.453	0.718	0.906	1.273	1.440	1.943	2.447	3.143	3.707	5.208
7	0.263	0.449	0.711	0.896	1.254	1.415	1.895	2.365	2.998	3.499	4.785
8	0.262	0.447	0.706	0.889	1.240	1.397	1.860	2.306	2.896	3.355	4.501
9	0.261	0.445	0.703	0.883	1.230	1.383	1.833	2.262	2.821	3.250	4.297
10	0.260	0.444	0.700	0.879	1.221	1.372	1.812	2.228	2.764	3.169	4.144
11	0.260	0.443	0.697	0.876	1.214	1.363	1.796	2.201	2.718	3.106	4.025
12	0.259	0.442	0.695	0.873	1.209	1.356	1.782	2.179	2.681	3.055	3.930
13	0.259	0.441	0.694	0.870	1.204	1.350	1.771	2.160	2.650	3.012	3.852
14	0.258	0.440	0.692	0.868	1.200	1.345	1.761	2.145	2.624	2.977	3.787
15	0.258	0.439	0.691	0.866	1.197	1.341	1.753	2.131	2.602	2.947	3.733
16	0.258	0.439	0.690	0.865	1.194	1.337	1.746	2.120	2.583	2.921	3.686
17	0.257	0.438	0.689	0.863	1.191	1.333	1.740	2.110	2.567	2.898	3.646
18	0.257	0.438	0.688	0.862	1.189	1.330	1.734	2.101	2.552	2.878	3.610
19	0.257	0.438	0.688	0.861	1.187	1.328	1.729	2.093	2.539	2.861	3.579
20	0.257	0.437	0.687	0.860	1.185	1.325	1.725	2.086	2.528	2.845	3.552
21	0.257	0.437	0.686	0.859	1.183	1.323	1.721	2.080	2.518	2.831	3.527
22	0.256	0.437	0.686	0.858	1.182	1.321	1.717	2.074	2.508	2.819	3.505
23	0.256	0.436	0.685	0.858	1.180	1.319	1.714	2.069	2.500	2.807	3.485
24	0.256	0.436	0.685	0.857	1.179	1.318	1.711	2.064	2.492	2.797	3.467
25	0.256	0.436	0.684	0.856	1.178	1.316	1.708	2.060	2.485	2.787	3.450
26	0.256	0.436	0.684	0.856	1.177	1.315	1.706	2.056	2.479	2.779	3.435
27	0.256	0.435	0.684	0.855	1.176	1.314	1.703	2.052	2.473	2.771	3.421
28	0.256	0.435	0.683	0.855	1.175	1.313	1.701	2.048	2.467	2.763	3.408
29	0.256	0.435	0.683	0.854	1.174	1.311	1.699	2.045	2.462	2.756	3.396
30	0.256	0.435	0.683	0.854	1.173	1.310	1.697	2.042	2.457	2.750	3.385
35	0.255	0.434	0.682	0.852	1.170	1.306	1.690	2.030	2.438	2.724	3.340
40	0.255	0.434	0.681	0.851	1.167	1.303	1.684	2.021	2.423	2.704	3.307
45	0.255	0.434	0.680	0.850	1.165	1.301	1.679	2.014	2.412	2.690	3.281
50	0.255	0.433	0.679	0.849	1.164	1.299	1.676	2.009	2.403	2.678	3.261
55	0.255	0.433	0.679	0.848	1.163	1.297	1.673	2.004	2.396	2.668	3.245
60	0.254	0.433	0.679	0.848	1.162	1.296	1.671	2.000	2.390	2.660	3.232
∞	0.253	0.431	0.674	0.842	1.150	1.282	1.645	1.960	2.326	2.576	3.090

NORMAL CUMULATIVE DISTRIBUTION FUNCTION

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000